

Practical - 30: Implementation of File, FileReader and FileWriter class using java

121) Write a program that counts number of characters, words, and lines in a file. Use exceptions to check whether the file that is read exists or not. (A)

```
import java.io.BufferedReader;
import java.io.FileReader;
import java.io.FileNotFoundException;
import java.io.IOException;

public class p121 {
    public static void main(String[] args) {

        int chars = 0, words = 0, lines = 0;

        try {
            BufferedReader br = new BufferedReader(
                new FileReader("sample.txt"));

            String line;

            while ((line = br.readLine()) != null) {
                lines++;

                chars += line.length();

                String arr[] = line.trim().split("\\s+");

                if (!line.trim().isEmpty())
                    words += arr.length;
            }

            br.close();

            System.out.println("Lines = " + lines);
            System.out.println("Words = " + words);
            System.out.println("Characters = " + chars);

        } catch (FileNotFoundException e) {
            System.out.println("File does not exist.");
        } catch (IOException e) {
            System.out.println(e);
        }
    }
}
```

122) Write a program to replace all “word1” by “word2” from a file1, and output is written to file2 file and display the no. of replacement. (A)

```
import java.io.*;
```

```
public class p122 {
    public static void main(String[] args) {

        int count = 0;

        try {
            BufferedReader br = new BufferedReader(
                new FileReader("file1.txt"));

            BufferedWriter bw = new BufferedWriter(
                new FileWriter("file2.txt"));

            String line;

            while ((line = br.readLine()) != null) {

                String oldLine = line;

                line = line.replace("word1", "word2");

                int diff = (oldLine.length()
                    - oldLine.replace("word1", "").length())
                    / "word1".length();

                count += diff;

                bw.write(line);
                bw.newLine();
            }

            br.close();
            bw.close();

            System.out.println("Number of Replacements = " + count);

        } catch (IOException e) {
            System.out.println(e);
        }
    }
}
```

123) Write an application that reads a file and counts the number of occurrences of digit 5. Supply the file name as a commandline argument. (B)

```
import java.io.FileInputStream;
import java.io.IOException;

public class p123 {
    public static void main(String[] args) {

        if (args.length == 0) {
```

```

        System.out.println("Provide file name.");
        return;
    }

    int count = 0;

    try {
        FileInputStream fis =
            new FileInputStream(args[0]);

        int ch;

        while ((ch = fis.read()) != -1) {

            if ((char) ch == '5')
                count++;

        }

        fis.close();

        System.out.println(
            "Occurrences of digit 5 = " + count);

    } catch (IOException e) {
        System.out.println(e);
    }
}
}

```

124) Create a class called Student. Write a student manager program to manipulate the student information from files by using FileInputStream and FileOutputStream. (C)

```

import java.io.*;

class Student {
    int rollNo;
    String name;

    Student(int rollNo, String name) {
        this.rollNo = rollNo;
        this.name = name;
    }
}

public class p124 {
    public static void main(String[] args) {

        Student s1 = new Student(101, "Aniruddh");
        Student s2 = new Student(102, "Rahul");
    }
}

```

```
try {  
  
    FileOutputStream fos =  
        new FileOutputStream("student.txt");  
  
    String data =  
        s1.rollNo + " " + s1.name + "\n" +  
        s2.rollNo + " " + s2.name;  
  
    fos.write(data.getBytes());  
    fos.close();  
  
    System.out.println("Student data stored.");  
  
    FileInputStream fis =  
        new FileInputStream("student.txt");  
  
    int ch;  
  
    System.out.println("\nStudent Records:");  
  
    while ((ch = fis.read()) != -1) {  
        System.out.print((char) ch);  
    }  
  
    fis.close();  
  
} catch (IOException e) {  
    System.out.println(e);  
}  
}
```